

Milestone 3 Report

CSCE3701 - Software Engineering (Fall 2025)

Hospital Management System - Carevo

Group Number: 1

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Sprint Summary

- **Sprint Goal:**

The goal of Sprint 1 was to establish the foundational layer of the Carevo hospital management mobile application. This included implementing the core authentication system (Sign-Up and Login), setting up the patient main page, displaying clinic categories, enabling appointment booking, cancelling, and creating the navigation tabs required for the patient view. This sprint ensures that the essential entry points and primary patient workflows are functional.

- **Scope Overview:**

The user stories tackled in this sprint were:

- 1) As a system user, I want to access a login page where I can enter my email and password so that I can securely sign in to my account.
- 2) As a patient, I want to create a new account by providing my personal details and contact information so that I can access the hospital's online services securely.
- 3) As a patient, I want to access a main page that has my info, so that I can provide the needed info to the hospital.
- 4) As a patient, I want to view all clinics by category so that I can choose one to book for.
- 5) As a patient, I want to book appointments so that I can visit my doctor at a convenient time.
- 6) As a patient, I want to cancel my booked appointments so that I can free up the slot for others.

- **Expected Deliverable:**

At the end of Sprint 1, the following components of the MVP are now functional:

- Fully functional Sign-Up system
- Working Login system
- Patient main page UI connected to user data
- Clinics listing by category
- Appointment booking workflow (clinic → doctor → time → confirm)
- Appointment cancelling workflow (Appointments → Cancel → Yes → Ok)
- Initial navigation tabs for patient view

These features form the core backbone of our Carevo app and enable the patient user journey from registration → login → navigation → clinic browsing, → appointment booking.

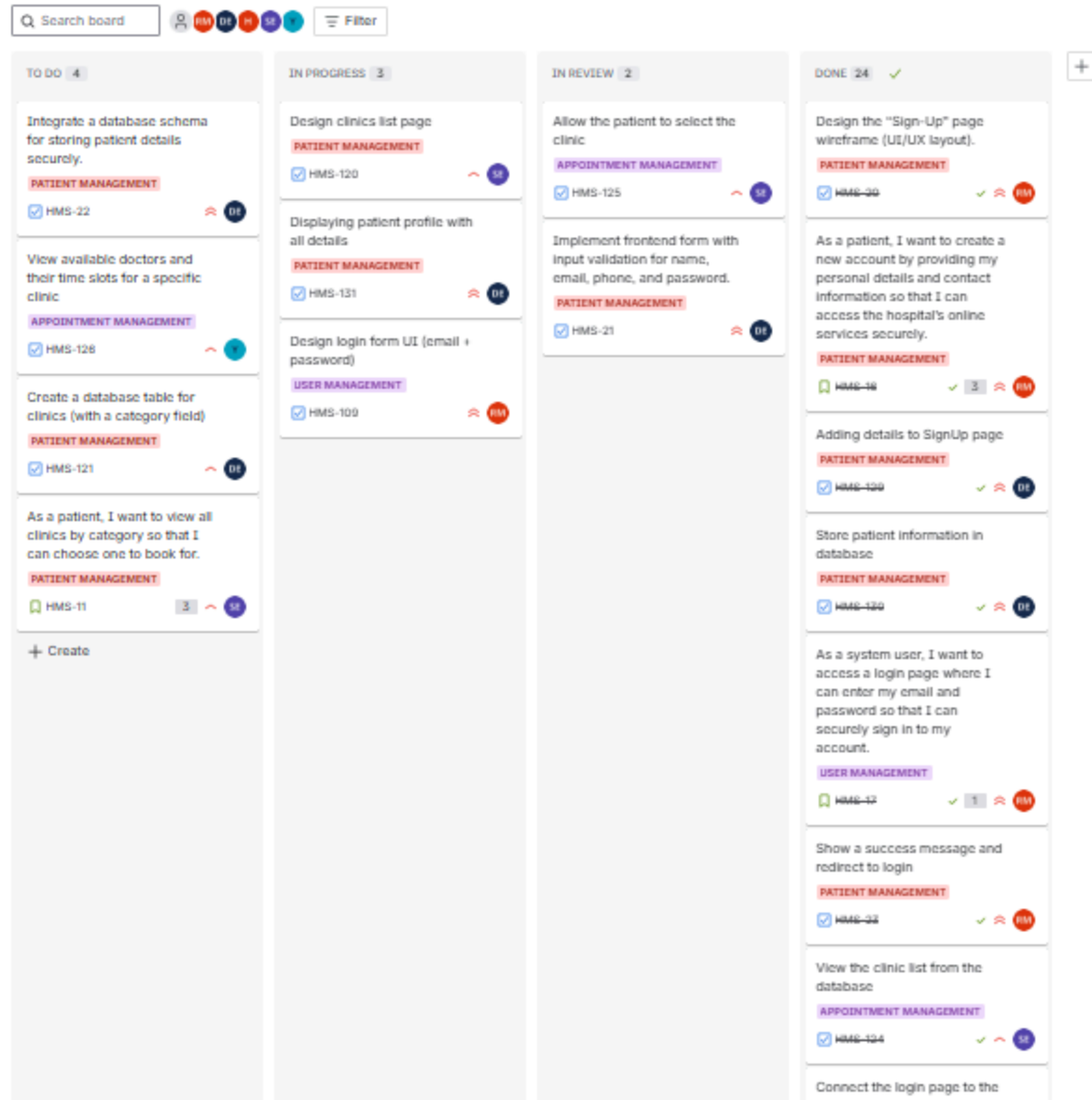
Sprint Board Snapshot

Jira Account:

<https://aucegypt-team-tk4cf28k.atlassian.net/jira/software/projects/HMS/boards/1>

The first snapshot was taken during our sprint before completing all the tasks. Thus, some of them are done, in progress, or in review.

To build the core foundation of the Hospital Management System, focusing on patient access, authentication, and appointment booking flow.			
☐ HMS-106 As a patient, I want to create a new account by providing my personal details and contact information so that I can access the hospital's online services securely.	PATIENT MANAGEMENT	DONE	100%
☐ HMS-100 Design the "Sign-Up" page wireframe (UI/UX layout).	PATIENT MANAGEMENT	DONE	100%
☐ HMS-109 Design login form UI (email + password)	USER MANAGEMENT	DONE	100%
☐ HMS-109 Adding details to Signup page	PATIENT MANAGEMENT	DONE	100%
☐ HMS-100 Design clinics list page	PATIENT MANAGEMENT	DONE	100%
☐ HMS-100 Store patient information in database	PATIENT MANAGEMENT	DONE	100%
☐ HMS-103 Displaying patient profile with all details	PATIENT MANAGEMENT	DONE	100%
☐ HMS-105 Allow the patient to select the clinic	APPOINTMENT MANAGEMENT	DONE	100%
☐ HMS-106 View available doctors and their time slots for a specific clinic	APPOINTMENT MANAGEMENT	DONE	100%
☐ HMS-25 Implement frontend form with input validation for name, email, phone, and password.	PATIENT MANAGEMENT	DONE	100%
☐ HMS-29 Integrate a database schema for storing patient details securely.	PATIENT MANAGEMENT	DONE	100%
☐ HMS-29 Show a success message and redirect to login	PATIENT MANAGEMENT	DONE	100%
☐ HMS-107 As a system user, I want to access a login page where I can enter my email and password so that I can securely sign in to my account.	USER MANAGEMENT	DONE	100%
☐ HMS-124 View the clinic list from the database	APPOINTMENT MANAGEMENT	IN REVIEW	100%
☐ HMS-115 Connect the login page to the user database	USER MANAGEMENT	IN REVIEW	100%
☐ HMS-112 Show error message for wrong credential	USER MANAGEMENT	IN REVIEW	100%
☐ HMS-109 As a patient, I want to access a main page that has my info, so that I can provide the needed info to the hospital.	PATIENT MANAGEMENT	DONE	100%
☐ HMS-106 Design main page layout (Basic patient info)	PATIENT MANAGEMENT	DONE	100%
☐ HMS-107 Display services by adding navigation buttons	PATIENT MANAGEMENT	DONE	100%
☐ HMS-109 Fetch and display logged-in user details	PATIENT MANAGEMENT	DONE	100%
☐ HMS-11 As a patient, I want to view all clinics by category so that I can choose one to book for.	PATIENT MANAGEMENT	IN REVIEW	100%
☐ HMS-121 Create a database table for clinics (with a category field)	PATIENT MANAGEMENT	IN REVIEW	100%
☐ HMS-109 Display list of clinics with basic info (name, specialty, availability)	PATIENT MANAGEMENT	DONE	100%
☐ HMS-7 As a patient, I want to book an appointment with a doctor so that I can receive a consultation.	APPOINTMENT MANAGEMENT	DONE	100%
☐ HMS-103 Design appointment page (UI/UX layout)	APPOINTMENT MANAGEMENT	DONE	100%
☐ HMS-109 Display confirmation message	APPOINTMENT MANAGEMENT	DONE	100%
☐ HMS-109 The appointment must be added to the patient's appointments in the database	APPOINTMENT MANAGEMENT	DONE	100%
☐ HMS-102 Booked appointment reflected in upcoming appointments	APPOINTMENT MANAGEMENT	DONE	100%
☐ HMS-12 As a patient, I want to view all my upcoming and past appointments so that I can keep track of my visits.	APPOINTMENT MANAGEMENT	IN REVIEW	100%
☐ HMS-133 View cancelled appointments	APPOINTMENT MANAGEMENT	IN REVIEW	100%
☐ HMS-134 View upcoming appointments	APPOINTMENT MANAGEMENT	IN REVIEW	100%
☐ HMS-136 Integrate with database	APPOINTMENT MANAGEMENT	IN REVIEW	100%
☐ HMS-9 As a patient, I want to cancel an appointment so that I can release the slot if I cannot attend.	APPOINTMENT MANAGEMENT	IN PROGRESS	100%



The following snapshot was taken after completing the entire sprint and achieving the goal of the sprint by completing all the tasks.

To build the core foundation of the Hospital Management System, focusing on patient access, authentication, and appointment booking flow.			
☐ HMS-1-6 As a patient, I want to create a new account by providing my personal details and contact information so that I can access the hospital's online services securely.	PATIENT MANAGEMENT	DONE	OK
☐ HMS-1-7 Design the "Sign-Up" page wireframe (UI/UX layout).	PATIENT MANAGEMENT	DONE	OK
☐ HMS-1-8 Design login form UI (email + password)	USER MANAGEMENT	DONE	OK
☐ HMS-1-9 Adding details to SignUp page	PATIENT MANAGEMENT	DONE	OK
☐ HMS-1-10 Design clinics list page	PATIENT MANAGEMENT	DONE	OK
☐ HMS-1-11 Store patient information in database	PATIENT MANAGEMENT	DONE	OK
☐ HMS-1-12 Displaying patient profile with all details	PATIENT MANAGEMENT	DONE	OK
☐ HMS-1-13 Allow the patient to select the clinic	APPOINTMENT MANAGEMENT	DONE	OK
☐ HMS-1-14 View available doctors and their time slots for a specific clinic	APPOINTMENT MANAGEMENT	DONE	OK
☐ HMS-1-15 Implement frontend form with input validation for name, email, phone, and password.	PATIENT MANAGEMENT	DONE	OK
☐ HMS-1-16 Integrate a database schema for storing patient details securely.	PATIENT MANAGEMENT	DONE	OK
☐ HMS-1-17 Show a success message and redirect to login	PATIENT MANAGEMENT	DONE	OK
☐ HMS-1-18 As a system user, I want to access a login page where I can enter my email and password so that I can securely sign in to my account.	USER MANAGEMENT	DONE	OK
☐ HMS-1-19 View the clinic list from the database	APPOINTMENT MANAGEMENT	DONE	OK
☐ HMS-1-20 Connect the login page to the user database	USER MANAGEMENT	DONE	OK
☐ HMS-1-21 Show error message for wrong credential	USER MANAGEMENT	DONE	OK
☐ HMS-1-22 As a patient, I want to access a main page that has my info, so that I can provide the needed info to the hospital.	PATIENT MANAGEMENT	DONE	OK
☐ HMS-1-23 Design main page layout (basic patient info)	PATIENT MANAGEMENT	DONE	OK
☐ HMS-1-24 Display services by adding navigation buttons	PATIENT MANAGEMENT	DONE	OK
☐ HMS-1-25 Fetch and display logged-in user details	PATIENT MANAGEMENT	DONE	OK
☐ HMS-1-26 As a patient, I want to view all clinics by category so that I can choose one to book for.	PATIENT MANAGEMENT	DONE	OK
☐ HMS-1-27 Create a database table for clinics (with a category field)	PATIENT MANAGEMENT	DONE	OK
☐ HMS-1-28 Display list of clinics with basic info (name, specialty, availability)	PATIENT MANAGEMENT	DONE	OK
☐ HMS-1-29 As a patient, I want to book an appointment with a doctor so that I can receive a consultation.	APPOINTMENT MANAGEMENT	OK	OK
☐ HMS-1-30 Design appointment page (UI/UX layout)	APPOINTMENT MANAGEMENT	DONE	OK
☐ HMS-1-31 Display confirmation message	APPOINTMENT MANAGEMENT	DONE	OK
☐ HMS-1-32 The appointment must be added to the patient's appointments in the database	APPOINTMENT MANAGEMENT	DONE	OK
☐ HMS-1-33 Booked appointment reflected in upcoming appointments	APPOINTMENT MANAGEMENT	DONE	OK
☐ HMS-1-34 As a patient, I want to view all my upcoming and past appointments so that I can keep track of my visits.	APPOINTMENT MANAGEMENT	OK	OK
☐ HMS-1-35 View cancelled appointments	APPOINTMENT MANAGEMENT	OK	OK
☐ HMS-1-36 view upcoming appointments	APPOINTMENT MANAGEMENT	OK	OK
☐ HMS-1-37 Integrate with database	APPOINTMENT MANAGEMENT	OK	OK
☐ HMS-1-38 As a patient, I want to cancel an appointment so that I can release the slot if I cannot attend.	APPOINTMENT MANAGEMENT	OK	OK

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IN REVIEW

DONE 33

As a patient, I want to create a new account by providing my personal details and contact information so that I can access the hospital's online services securely.

PATIENT MANAGEMENT

HMS-1-6

Design the "Sign-Up" page wireframe (UI/UX layout).

PATIENT MANAGEMENT

HMS-1-7

Design login form UI (email + password)

USER MANAGEMENT

HMS-1-8

Adding details to SignUp page

PATIENT MANAGEMENT

HMS-1-9

Design clinics list page

PATIENT MANAGEMENT

HMS-1-10

Store patient information in database

PATIENT MANAGEMENT

HMS-1-11

Displaying patient profile with all details

PATIENT MANAGEMENT

HMS-1-12

Allow the patient to select the clinic

APPOINTMENT MANAGEMENT

HMS-1-13

Give feedback on the n...

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TO DO

IN PROGRESS

IN REVIEW

DONE 33

View available doctors and their time slots for a specific clinic.

APPOINTMENT MANAGEMENT

HMAS-126

Implement frontend form with input validation for name, email, phone, and password.

PATIENT MANAGEMENT

HMAS-24

Integrate a database schema for storing patient details securely.

PATIENT MANAGEMENT

HMAS-22

Show a success message and redirect to login

PATIENT MANAGEMENT

HMAS-25

As a system user, I want to access a login page where I can enter my email and password so that I can securely sign in to my account.

USER MANAGEMENT

HMAS-12

View the clinic list from the database

APPOINTMENT MANAGEMENT

HMAS-124

Connect the login page to the user database

USER MANAGEMENT

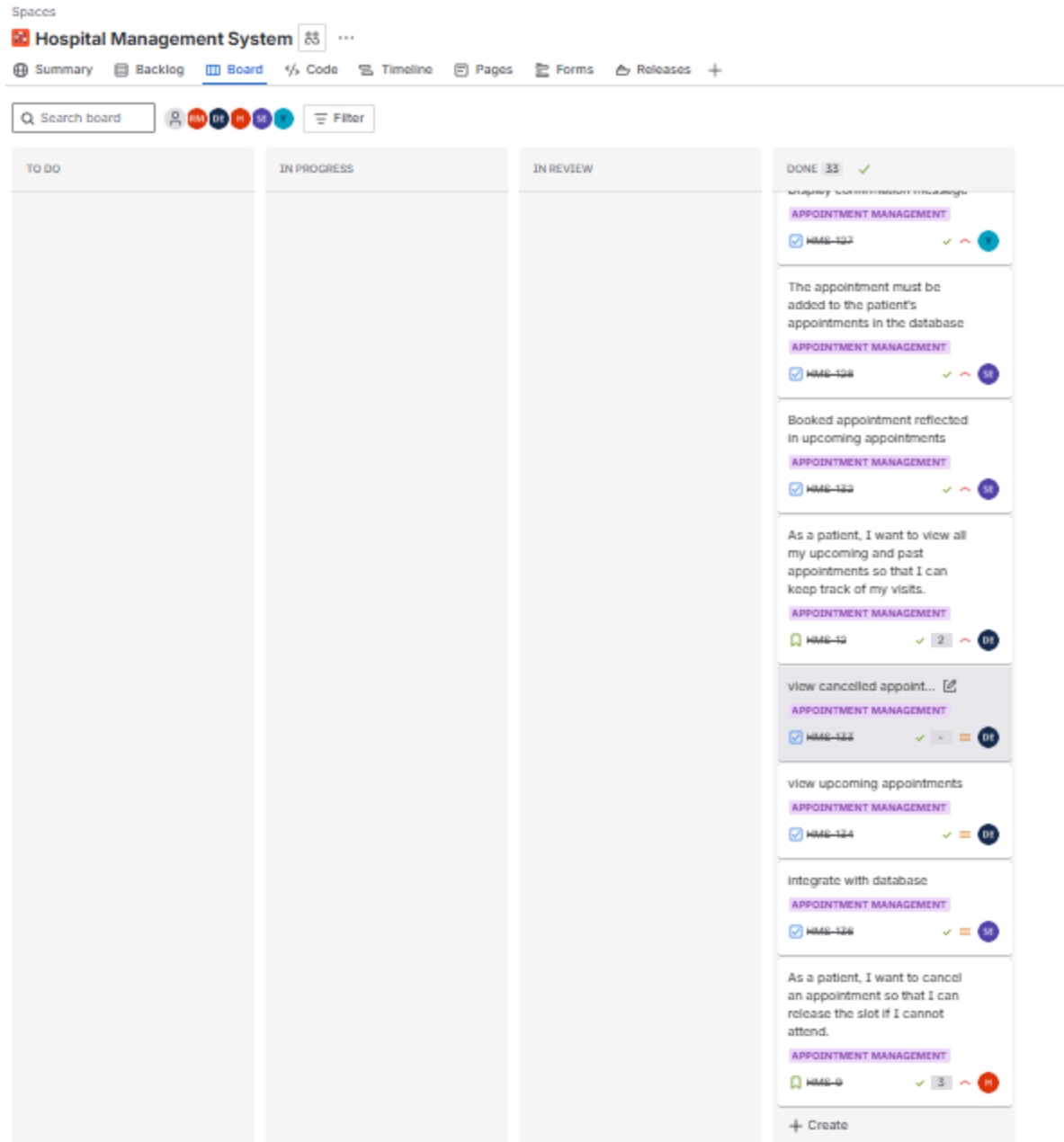
HMAS-116

Show error message for wrong credential

Give feedback on the n...

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TO DO	IN PROGRESS	IN REVIEW	DONE 33
			USER MANAGEMENT ☒ HME-116 Show error message for wrong credential USER MANAGEMENT ☒ HME-115 As a patient, I want to access a main page that has my info, so that I can provide the needed info to the hospital. PATIENT MANAGEMENT HME-119 3 Design main page layout (basic patient info) PATIENT MANAGEMENT ☒ HME-116 Display services by adding navigation buttons PATIENT MANAGEMENT ☒ HME-117 Fetch and display logged-in user details PATIENT MANAGEMENT ☒ HME-119 As a patient, I want to view all clinics by category so that I can choose one to book for. PATIENT MANAGEMENT HME-11 3 Create a database table for clinics (with a category field) PATIENT MANAGEMENT ☒ HME-124 Display list of clinics with basic info (name, specialty,



For *To Do* items, we discuss which to assign to each person. When someone starts working on their task, they mark them as *In Progress*. When they're done according to our DoD, they put them in the *In Review* section. Once someone else checks the task and verifies it's working correctly, they can put it in the *Done* section.

Working Software Demo

<https://youtu.be/Si4ctQFg4Bk>

Code Evidence

Repository link:

<https://github.com/Rawan10101/Carevo.git>

Each team member should have at least one commit merged this sprint:

Each team member contributed code during Sprint 1. Individual commits are visible in the repository history, covering the Sign-Up system, Login system, patient main page, clinics listing, appointment booking, and other features.

Branch Conventions and Merge Activity:

- Most work for Sprint 1 was pushed directly to the **main branch**.
- **Booking Branch:** Features for canceling and rescheduling appointments were developed on a separate branch named booking and later **merged into main**.
- **Doctors/Clinics Branch:** Work related to handling doctors, clinics, and appointments was done on another branch and subsequently merged into main.
- The team recognizes that using feature branches more consistently for all tasks will improve collaboration and reduce risks in future sprints.

Sprint Retrospective Notes

What went well

- **Clear Goal Alignment:** The team successfully implemented the planned core authentication features (Sign-Up and Login), patient main page, accessing clinics with categories, and booking flow.
- **Effective Team Collaboration:** Collaboration between frontend and backend developers was smooth; tasks were well divided and communicated.
- **Tool Efficiency:** Firebase integration for authentication and Firestore for patient and clinic data worked reliably and reduced backend complexity.
- **UI/UX Implementation:** Basic layouts for patient main page, clinic listing, and appointment booking were successfully translated into functional screens.
- **Testing and Debugging:** Early testing on devices allowed quick detection of input validation and navigation issues.

What didn't go well

- **Time Management:** Some tasks, such as integrating clinic listings with database queries, took longer than expected.

- Dependency on Firebase: Limited knowledge of Firestore queries led to slower implementation for complex features like appointment filtering.
- Incomplete Error Handling: Some edge cases for login and sign-up errors were not fully accounted for during testing.

Improvements for next sprint

1. Optimize Database Queries

Refactor Firestore queries to improve performance when fetching clinics and available appointments.

Owner: Backend Developer

Deadline: Mid Sprint 2

2. Enhance Error Handling

Cover additional validation edge cases for sign-up/login, including duplicate accounts and invalid input formats.

Owner: Frontend & Backend Devs

Deadline: Mid Sprint 2

3. Navigation Testing

Conduct thorough end-to-end navigation tests to ensure smooth patient journey.

Owner: Developers

Deadline: End Sprint 2

4. Time Management Practices

Assign clear owners to tasks with realistic deadlines.

Team Reflections & Roles

Team Roles:

Doha: Developer **Salma:** Developer **Yara:** Developer

Habeba: Developer **Rawan:** Scrum Master & Developer

If a client was available, we would assume the Product Owner role, providing requirements, feedback, and priority guidance. In this sprint, the team collectively handled that responsibility through Habeba.

Collaboration Dynamics:

The team collaborated closely throughout Sprint 1. Tasks were divided among all developers, with daily communication ensuring alignment on UI implementation, Firebase integration, and

patient workflow features. As Scrum Master, Rawan facilitated task tracking, progress updates, and coordination between team members.

Planned Adjustments for Sprint 2:

- Adopt a more consistent branching strategy for all features to improve code management and reduce conflicts.
- Schedule short, focused check-ins to identify blockers earlier and streamline workflow.

Burndown / Progress Chart

